

PROJECT SUMMARY

Specific problem my track addresses. Measuring and explaining HealthConnect's no-show problem: how large it is, where and when it concentrates, and which factors (reminders, lead time, history, distance, scheduling) are associated with it, so the clinic can target interventions and the other tracks can build on a clean evidence base.

Resources used. The appointment dataset (5,000 rows, 18 columns), the data dictionary (definitions and validation rules), the clinic knowledge base (cancellation/rescheduling procedures, to separate cancellations from no-shows), and the Week 4 brief.

Key observations from Week 4. The data covers roughly Jan 2025–Jun 2026 with one row per appointment and repeating patients. Three outcomes exist (Attended, No-Show, Cancelled) and an initial scan suggests around 4 in 10 appointments end as no-shows, a large, real problem. Distance and waiting time have limited missing values (as the dictionary warns); reminder_channel = None exactly when no reminder was sent; previous no-shows never exceed previous appointments. One issue needs fixing in Week 5: the dates arrive as M/D/YYYY although the dictionary describes ISO format.

Proposed approach. Week 5: clean and type the data in Power Query (without touching the original file), implement five proposed KPIs as DAX measures, build descriptive no-show visuals (by reminder/channel, lead-time band, distance band, time/day, type, month), and cross-check numbers in Excel before drawing conclusions.

Key considerations. Cancelled must be measured separately from No-Show; missing distance/waiting values must not be silently imputed; small subgroups are unreliable; the date-format fix must happen before any time analysis; findings are correlations, not causation. Dependencies include the dictionary, the knowledge base, Power BI access, and alignment with the ML and GenAI tracks.

Proposed focus for Week 5. Complete the Power Query cleaning, implement and verify the five KPIs, and produce the first no-show analysis visuals.